Information About the Performed Analysis

You chose to analyze the following complexity measures:

1. C_{len} (n traces)
2. $C_{\text{DT-\#}}$ (n variants)
3. $C_{\text{DT-\%}}$ (ratio variants per number of traces)
4. C_{mag} (n events)
5. $C_{\text{TL-min}}$ (trace len min)
6. $C_{\text{TL-max}}$ (trace len max)
7. $C_{\text{TL-mean}}$ (trace len mean)
8. $C_{\text{TL-med}}$ (trace len median)
9. $C_{\text{TL-mode}}$ (trace len mode)
10. $C_{\text{TL-dev}}$ (trace len std)
11. $C_{\text{TL-var}}$ (trace len variance)
12. $C_{\text{TL-Q1}}$ (trace len q1)
13. $C_{\text{TL-Q3}}$ (trace len q3)
14. $C_{\text{TL-IQR}}$ (trace len iqr)
15. $C_{\text{TL-g-mean}}$ (trace len geometric mean)
16. $C_{\text{TL-g-dev}}$ (trace len geometric std)
17. $C_{\text{TL-h-mean}}$ (trace len harmonic mean)
18. $C_{\text{TL-skew}}$ (trace len skewness)
19. $C_{\text{TL-kurt}}$ (trace len kurtosis)
20. $C_{\text{TL-cvar}}$ (trace len coefficient variation)
21. $C_{\text{TL-ent}}$ (trace len entropy)
22. $C_{\text{TL-hist1}}$ (trace len hist1)
23. $C_{\text{TL-hist2}}$ (trace len hist2)
24. $C_{\text{TL-hist3}}$ (trace len hist3)
25. $C_{\text{TL-hist4}}$ (trace len hist4)
26. $C_{\text{TL-hist5}}$ (trace len hist5)
27. $C_{\text{TL-hist6}}$ (trace len hist6)
28. $C_{\text{TL-hist7}}$ (trace len hist7)
29. $C_{\text{TL-hist8}}$ (trace len hist8)
30. $C_{\text{TL-hist9}}$ (trace len hist9)
31. $C_{\text{TL-hist10}}$ (trace len hist10)
32. $C_{\text{TL-hist-skew}}$ (trace len skewness hist)
33. $C_{\text{TL-hist-kurt}}$ (trace len kurtosis hist)
34. $C_{\text{FREQ-T-max-r}}$ (ratio most common variant)
35. $C_{\text{FREQ-T-max-1\%}}$ (ratio top 1 variants)
36. $C_{\text{FREQ-T-max-5\%}}$ (ratio top 5 variants)
37. $C_{\text{FREQ-T-max-10\%}}$ (ratio top 10 variants)
38. $C_{\text{FREQ-T-max-20\%}}$ (ratio top 20 variants)
39. $C_{\text{FREQ-T-max-50\%}}$ (ratio top 50 variants)
40. $C_{\text{FREQ-T-max-75\%}}$ (ratio top 75 variants)
41. $C_{\text{FREQ-T-mean}}$ (mean variant occurrence)
42. $C_{\text{FREQ-T-dev}}$ (std variant occurrence)
43. $C_{\text{FREQ-T-skew}}$ (skewness variant occurrence)
44. $C_{\text{FREQ-T-kurt}}$ (kurtosis variant occurrence)
45. $C_{\text{DT-cov}}$ (coverage variants)
46. $C_{\text{DT-cov-\%}}$ (rel coverage variants)
47. $C_{\text{DT-het}}$ (heterogeneity rate variants)
48. $C_{\text{DT-sim}}$ (similarity rate variants)
49. C_{var} (n unique activities)
50. $C_{\text{FREQ-A-min}}$ (activities min)
51. $C_{\text{FREQ-A-max}}$ (activities max)
52. $C_{\text{FREQ-A-mean}}$ (activities mean)
53. $C_{\text{FREQ-A-med}}$ (activities median)
54. $C_{\text{FREQ-A-dev}}$ (activities std)
55. $C_{\text{FREQ-A-var}}$ (activities variance)

56. $C_{\text{FREQ-A-Q1}}$ (activities q1)
57. $C_{\text{FREQ-A-Q3}}$ (activities q3)
58. $C_{\text{FREQ-A-IRQ}}$ (activities iqr)
59. $C_{\text{FREQ-A-skew}}$ (activities skewness)
60. $C_{\text{FREQ-A-kurt}}$ (activities kurtosis)
61. $C_{\text{A-start-\#}}$ (n unique start activities)
62. $C_{\text{A-start-min}}$ (start activities min)
63. $C_{\text{A-start-max}}$ (start activities max)
64. $C_{\text{A-start-mean}}$ (start activities mean)
65. $C_{\text{A-start-med}}$ (start activities median)
66. $C_{\text{A-start-dev}}$ (start activities std)
67. $C_{\text{A-start-var}}$ (start activities variance)
68. $C_{\text{A-start-Q1}}$ (start activities q1)
69. $C_{\text{A-start-Q3}}$ (start activities q3)
70. $C_{\text{A-start-IQR}}$ (start activities iqr)
71. $C_{\text{A-start-skew}}$ (start activities skewness)
72. $C_{\text{A-start-kurt}}$ (start activities kurtosis)
73. $C_{\text{A-start-rel}}$ (rel unique start activities)
74. $C_{\text{A-end-\#}}$ (n unique end activities)
75. $C_{\text{A-end-min}}$ (end activities min)
76. $C_{\text{A-end-max}}$ (end activities max)
77. $C_{\text{A-end-mean}}$ (end activities mean)
78. $C_{\text{A-end-med}}$ (end activities median)
79. $C_{\text{A-end-dev}}$ (end activities std)
80. $C_{\text{A-end-var}}$ (end activities variance)
81. $C_{\text{A-end-Q1}}$ (end activities q1)
82. $C_{\text{A-end-Q3}}$ (end activities q3)
83. $C_{\text{A-end-IQR}}$ (end activities iqr)
84. $C_{\text{A-end-skew}}$ (end activities skewness)
85. $C_{\text{A-end-kurt}}$ (end activities kurtosis)
86. $C_{\text{A-end-rel}}$ (rel unique end activities)
87. $C_{\text{ENT-trace}}$ (eventropy trace)
88. $C_{\text{ENT-pref}}$ (eventropy prefix)
89. $C_{\text{ENT-pref-F}}$ (eventropy prefix flattened)
90. $C_{\text{ENT-LZ}}$ (eventropy lempel ziv)
91. $C_{\text{ENT-LZ-F}}$ (eventropy lempel ziv flattened)
92. $C_{\text{ENT-k-block-d1}}$ (eventropy k block diff 1)
93. $C_{\text{ENT-k-block-d3}}$ (eventropy k block diff 3)
94. $C_{\text{ENT-k-block-d5}}$ (eventropy k block diff 5)
95. $C_{\text{ENT-k-block-r1}}$ (eventropy k block ratio 1)
96. $C_{\text{ENT-k-block-r3}}$ (eventropy k block ratio 3)
97. $C_{\text{ENT-k-block-r5}}$ (eventropy k block ratio 5)
98. $C_{\text{ENT-knn-3}}$ (eventropy knn 3)
99. $C_{\text{ENT-knn-5}}$ (eventropy knn 5)
100. $C_{\text{ENT-knn-7}}$ (eventropy knn 7)
101. $C_{\text{epa-var}}$ (epa variant entropy)
102. $C_{\text{epa-n-var}}$ (epa normalized variant entropy)
103. $C_{\text{epa-seq}}$ (epa sequence entropy)
104. $C_{\text{epa-n-seq}}$ (epa normalized sequence entropy)
105. $C_{\text{epa-seq-L}}$ (epa sequence entropy linear forgetting)
106. $C_{\text{epa-n-seq-L}}$ (epa normalized sequence entropy linear forgetting)
107. $C_{\text{epa-seq-E}}$ (epa sequence entropy exponential forgetting)
108. $C_{\text{epa-n-seq-E}}$ (epa normalized sequence entropy exponential forgetting)
109. $C_{\text{s-T-div}}$ (simple trace diversity)
110. $C_{\text{a-T-div}}$ (advanced trace diversity)
111. $C_{\text{DA-min}}$ (distinct activities min)
112. $C_{\text{DA-max}}$ (distinct activities max)
113. $C_{\text{DA-avg}}$ (distinct activities mean)

114. $C_{\text{DA-dev}}$ (distinct activities std)
115. $C_{\text{DA-dens}}$ (event density)
116. $C_{\text{DA-ovrlp}}$ (distinct activities non overlap)
117. $C_{\text{comp-fact}}$ (complexity factor)
118. $C_{\text{loop-\#}}$ (n traces with loop)
119. $C_{\text{loop-rel-\#}}$ (avg traces with loop)
120. $C_{\text{loop-avg-\#}}$ (avg loops per trace)
121. $C_{\text{loop-max-\#}}$ (max loops per trace)
122. $C_{\text{loop-avg-size}}$ (avg loop size per trace)
123. $C_{\text{loop-max-size}}$ (max loop size per trace)
124. $C_{\text{rep-\#}}$ (n traces with repetition)
125. $C_{\text{rep-rel-\#}}$ (avg traces with repetition)
126. C_{succ} (number of successions)
127. C_{ties} (number of ties)
128. C_{struct} (structure)
129. C_{affinity} (average affinity)
130. C_{LZ} (lempel ziv complexity)
131. $C_{\text{dev-R}}$ (deviation from random)
132. $C_{\text{avg-dist}}$ (average edit distance)
133. $C_{\text{DFG-V}}$ (n nodes dfg)
134. $C_{\text{DFG-E}}$ (n edges dfg)
135. $C_{\text{DFG-cnc}}$ (coeff of connectivity dfg)
136. $C_{\text{DFG-adeg}}$ (avg node degree dfg)
137. $C_{\text{DFG-mdeg}}$ (max node degree dfg)
138. $C_{\text{DFG-dens}}$ (density dfg)
139. $C_{\text{DFG-struct}}$ (structure dfg)
140. $C_{\text{DFG-CN}}$ (cyclomatic number dfg)
141. $C_{\text{DFG-sep-\#}}$ (n cut vertices dfg)
142. $C_{\text{DFG-sep-r}}$ (separability ratio dfg)
143. $C_{\text{DFG-seq-r}}$ (sequentiality ratio dfg)
144. $C_{\text{DFG-cyc}}$ (cyclicity dfg)

Out of these measures, the following were excluded because they returned non-numeric values:

1. $C_{\text{A-start-skew}}$ (start activities skewness)
2. $C_{\text{A-start-kurt}}$ (start activities kurtosis)
3. $C_{\text{A-end-skew}}$ (end activities skewness)
4. $C_{\text{A-end-kurt}}$ (end activities kurtosis)
5. $C_{\text{TL-g-dev}}$ (trace len geometric std)
6. $C_{\text{TL-skew}}$ (trace len skewness)
7. $C_{\text{TL-kurt}}$ (trace len kurtosis)
8. $C_{\text{FREQ-T-skew}}$ (skewness variant occurrence)
9. $C_{\text{FREQ-T-kurt}}$ (kurtosis variant occurrence)
10. $C_{\text{FREQ-A-skew}}$ (activities skewness)
11. $C_{\text{FREQ-A-kurt}}$ (activities kurtosis)
12. $C_{\text{DT-sim}}$ (similarity rate variants)
13. $C_{\text{comp-fact}}$ (complexity factor)
14. $C_{\text{a-T-div}}$ (advanced trace diversity)

Furthermore, the following measures were excluded because their variance is 0 in the analyzed data:

1. $C_{\text{ENT-knn-3}}$ (eventropy knn 3)
2. $C_{\text{ENT-knn-5}}$ (eventropy knn 5)
3. $C_{\text{ENT-knn-7}}$ (eventropy knn 7)

Additionally, the following measures were excluded because they are correlated with other measures:

1. $C_{\text{DT-cov}}$ (coverage variants)
2. $C_{\text{DFG-V}}$ (n nodes dfg)
3. $C_{\text{DT-cov-\%}}$ (rel coverage variants)
4. $C_{\text{DFG-adeg}}$ (avg node degree dfg)

5. $C_{\text{ENT-k-block-r1}}$ (eventropy k block ratio 1)
6. $C_{\text{A-start-Q3}}$ (start activities q3)
7. $C_{\text{A-end-Q3}}$ (end activities q3)
8. $C_{\text{DFG-E}}$ (n edges dfg)
9. $C_{\text{TL-Q3}}$ (trace len q3)
10. $C_{\text{FREQ-A-Q3}}$ (activities q3)
11. $C_{\text{DFG-cnc}}$ (coeff of connectivity dfg)
12. $C_{\text{DFG-V}}$ (n nodes dfg)
13. $C_{\text{ENT-k-block-r1}}$ (eventropy k block ratio 1)

This leaves us with a total of 114 log measures to analyze.

Validity of the Factor Analysis

Your sample size in this analysis was 1450. It is recommended to have a sample size of at least $5 \cdot 144 = 720$. Bartlett's test of sphericity returned a χ^2 value of 564464.8990453014 and a p-value of 0.0. This means, the variables are probably correlated and we can perform a factor analysis.
The measure of sampling adequacy (MSA) returned the following results:

- C_{len}: 0.9182
- C_{var}: 0.9258
- C_{mag}: 0.9608
- $C_{\text{A-start-rel}}$: 0.868
- $C_{\text{A-start-min}}$: 0.8361
- $C_{\text{A-start-max}}$: 0.8235
- $C_{\text{A-start-mean}}$: 0.7739
- $C_{\text{A-start-med}}$: 0.8775
- $C_{\text{A-start-dev}}$: 0.8639
- $C_{\text{A-start-var}}$: 0.9406
- $C_{\text{A-start-Q1}}$: 0.8216
- $C_{\text{A-start-IQR}}$: 0.7645
- $C_{\text{A-end-\#}}$: 0.7423
- $C_{\text{A-end-rel}}$: 0.8812
- $C_{\text{A-end-min}}$: 0.8443
- $C_{\text{A-end-max}}$: 0.8177
- $C_{\text{A-end-mean}}$: 0.7608
- $C_{\text{A-end-med}}$: 0.8833
- $C_{\text{A-end-dev}}$: 0.8521
- $C_{\text{A-end-var}}$: 0.915
- $C_{\text{A-end-Q1}}$: 0.7915
- $C_{\text{A-end-IQR}}$: 0.6659
- $C_{\text{TL-min}}$: 0.8286
- $C_{\text{TL-max}}$: 0.9619
- $C_{\text{TL-mean}}$: 0.922
- $C_{\text{TL-med}}$: 0.9383
- $C_{\text{TL-mode}}$: 0.9487
- $C_{\text{TL-dev}}$: 0.9192
- $C_{\text{TL-var}}$: 0.8917
- $C_{\text{TL-Q1}}$: 0.9485
- $C_{\text{TL-IQR}}$: 0.9384
- $C_{\text{TL-g-mean}}$: 0.9281
- $C_{\text{TL-h-mean}}$: 0.9143
- $C_{\text{TL-cvar}}$: 0.8223
- $C_{\text{TL-ent}}$: 0.8357
- $C_{\text{TL-hist1}}$: 0.7795
- $C_{\text{TL-hist2}}$: 0.8532
- $C_{\text{TL-hist3}}$: 0.8342
- $C_{\text{TL-hist5}}$: 0.8298
- $C_{\text{TL-hist6}}$: 0.7169

- $C_{\text{TL-hist8}}$: 0.7781
- $C_{\text{TL-hist9}}$: 0.8009
- $C_{\text{TL-hist10}}$: 0.7287
- $C_{\text{TL-hist-skew}}$: 0.7934
- $C_{\text{TL-hist-kurt}}$: 0.7994
- $C_{\text{FREQ-T-max-r}}$: 0.9427
- $C_{\text{FREQ-T-max-1\%}}$: 0.8174
- $C_{\text{FREQ-T-max-5\%}}$: 0.9059
- $C_{\text{FREQ-T-max-10\%}}$: 0.9034
- $C_{\text{FREQ-T-max-20\%}}$: 0.9052
- $C_{\text{FREQ-T-max-50\%}}$: 0.9067
- $C_{\text{FREQ-T-max-75\%}}$: 0.891
- $C_{\text{FREQ-T-mean}}$: 0.8433
- $C_{\text{FREQ-T-dev}}$: 0.895
- $C_{\text{FREQ-A-min}}$: 0.8713
- $C_{\text{FREQ-A-max}}$: 0.9537
- $C_{\text{FREQ-A-mean}}$: 0.9115
- $C_{\text{FREQ-A-med}}$: 0.9481
- $C_{\text{FREQ-A-dev}}$: 0.9095
- $C_{\text{FREQ-A-var}}$: 0.9378
- $C_{\text{FREQ-A-Q1}}$: 0.9211
- $C_{\text{FREQ-A-IRQ}}$: 0.9505
- $C_{\text{DT-\#}}$: 0.9445
- $C_{\text{DT-\%}}$: 0.9233
- $C_{\text{DT-het}}$: 0.8911
- $C_{\text{DA-min}}$: 0.808
- $C_{\text{DA-max}}$: 0.9625
- $C_{\text{DA-avg}}$: 0.9285
- $C_{\text{DA-dev}}$: 0.9154
- $C_{\text{DA-dens}}$: 0.9279
- $C_{\text{DA-ovrlp}}$: 0.8072
- $C_{\text{s-T-div}}$: 0.8294
- $C_{\text{loop-\#}}$: 0.9463
- $C_{\text{loop-rel-\#}}$: 0.9231
- $C_{\text{loop-avg-\#}}$: 0.867
- $C_{\text{loop-max-\#}}$: 0.9264
- $C_{\text{loop-avg-size}}$: 0.8893
- $C_{\text{loop-max-size}}$: 0.9121
- $C_{\text{rep-\#}}$: 0.955
- $C_{\text{rep-rel-\#}}$: 0.9337
- $C_{\text{DFG-mdeg}}$: 0.9483
- $C_{\text{DFG-dens}}$: 0.8583
- $C_{\text{DFG-struct}}$: 0.872
- $C_{\text{DFG-CN}}$: 0.8355
- $C_{\text{DFG-sep-\#}}$: 0.8217
- $C_{\text{DFG-sep-r}}$: 0.8314
- $C_{\text{DFG-seq-r}}$: 0.9389
- $C_{\text{DFG-cyc}}$: 0.9501
- C_{succ}: 0.8432
- C_{ties}: 0.8742
- C_{struct}: 0.7598
- C_{affinity}: 0.8834
- C_{LZ}: 0.9304
- $C_{\text{dev-R}}$: 0.9376
- $C_{\text{avg-dist}}$: 0.9331
- $C_{\text{epa-var}}$: 0.9259
- $C_{\text{epa-n-var}}$: 0.863
- $C_{\text{epa-seq}}$: 0.8935

- $C_{\text{epa-n-seq}}$: 0.8869
- $C_{\text{epa-seq-L}}$: 0.8965
- $C_{\text{epa-n-seq-L}}$: 0.881
- $C_{\text{epa-seq-E}}$: 0.8943
- $C_{\text{epa-n-seq-E}}$: 0.882
- $C_{\text{ENT-trace}}$: 0.9492
- $C_{\text{ENT-pref}}$: 0.9497
- $C_{\text{ENT-pref-F}}$: 0.948
- $C_{\text{ENT-LZ}}$: 0.9563
- $C_{\text{ENT-LZ-F}}$: 0.9185
- $C_{\text{ENT-k-block-d1}}$: 0.8561
- $C_{\text{ENT-k-block-d3}}$: 0.9107
- $C_{\text{ENT-k-block-d5}}$: 0.946
- $C_{\text{ENT-k-block-r3}}$: 0.8937
- $C_{\text{ENT-k-block-r5}}$: 0.9507
- overall MSA: 0.8964956318283351

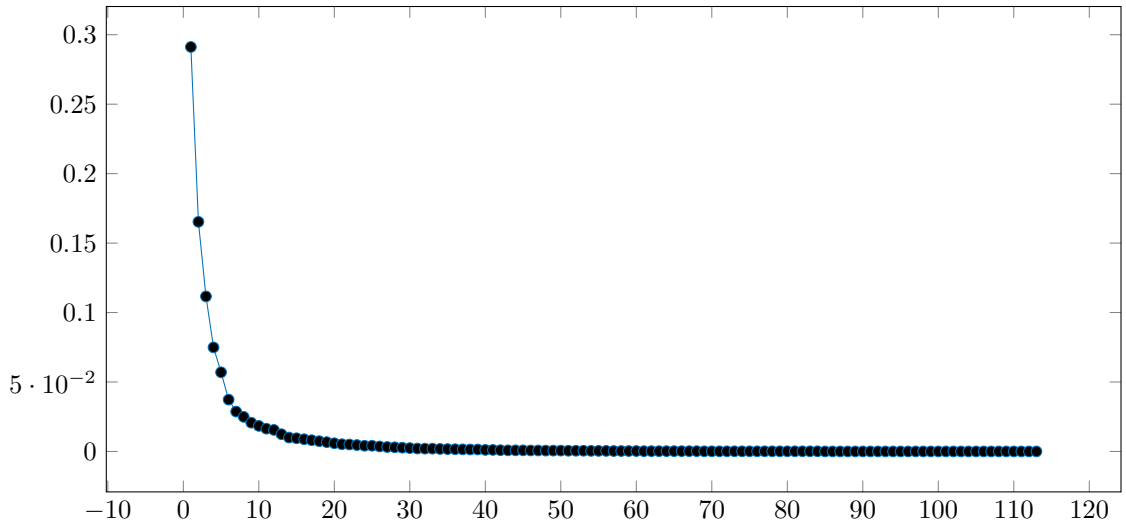
During the analysis, the MSA values of the following complexity measures were too low:

- $C_{\text{A-start-\#}}$: 0.5792
- $C_{\text{TL-hist4}}$: 0.565
- $C_{\text{TL-hist7}}$: 0.5204

These complexity measures were left out of the analysis to produce dependable results.

Chosen Number of Factors

The Scree test resulted in the following Eigenvalue plot:



You choose to set the number of expected factors to 10.